Impulse Response Descriptions for LTI Systems

Copyright © Barry Van Veen 2014

Feel free to pass this ebook around the web... but please do not modify any of its contents. Thanks!

AllSignalProcessing.com

Key Concepts

1) The impulse response of a system $h[n]$ is the output when the input is an impulse $\delta[n]$.

2) If the input $x[n]$ is expressed as a weighted sum of shifted impulses $\delta[n-l]$, the output $y[n]$ of an LTI system is expressed as a weighted sum of time-shifted impulse responses $h[n-l]$.

    a) The weights are the values of the input signal.

    b) The weighted sum of time-shifted impulse responses is called the *convolution* of the input with the impulse re-

sponse.

$$y[n] = x[n] * h[n] = \sum_{l=-\infty}^{\infty} x[l]h[n-l]$$

c) The convolution operation is commutative, so it can equivalently be written as

$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

3) The impulse response reflects key properties of the system.

a) The impulse response of a causal system is zero for times $n < 0$.

b) A finite impulse response (FIR) system has a finite number of non zero terms in the impulse response.

c) An infinite impulse response (IIR) system has an infinite number of non zero terms in the impulse response.

d) The impulse response of a cascade of systems is the convolution of the impulse responses of the systems in the cascade.

4) FIR systems are described by non recursive difference equations: the output depends only on the present/past values of the input.

5) IIR systems are described by recursive difference equations: the output depends on past values of the output.

6) The impulse response of a continuous-time system is a mathematical model for the response to an ideal, large amplitude, short duration input.

    a) If the input $x(t)$ is expressed as a weighted integral of shifted impulses $\delta(t-\tau)$, the output $y(t)$ of an LTI system is expressed as a weighted integral of time-shifted impulse responses $h(t-\tau)$.

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

or

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$$

b) The convolution integral and impulse response are used for analysis of continuous-time systems.

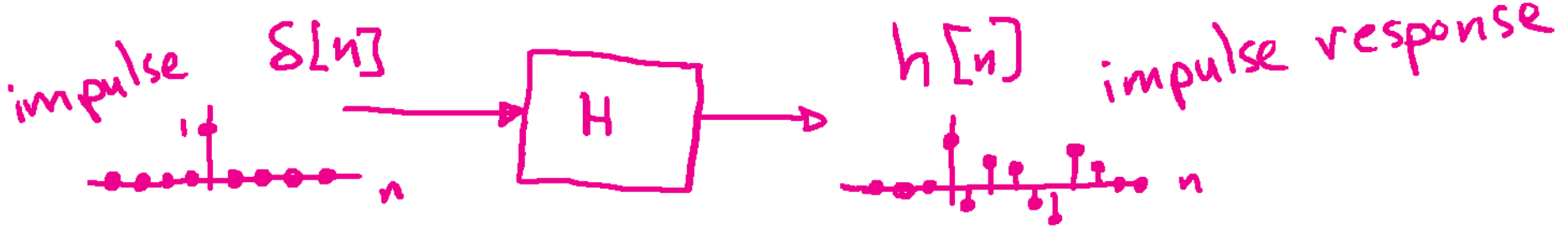
Video Screenshots

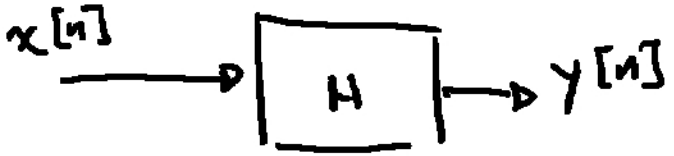
0:09

Characterizing linear, time-invariant systems using canonical inputs —

1) Write arbitrary input as a weighted sum of time-shifted canonical input

2) Output is a weighted sum of time-shifted canonical outputs





$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k] = h[n] * x[n]$$

"convolution"

operator notation

2:25

Properties:

1) Causal system

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\,x[n-k]$$

$$= \ldots + h[-2]x[n+2] + h[-1]x[n+1] + h[0]x[n] + h[1]x[n-1] + \ldots$$

Causal $\Leftrightarrow$ $h[n] = 0$ for $n < 0$

2) Finite impulse response (FIR): number of nonzero $h[n]$ is finite

Most common form $h[n] = 0$ $n<0$, $n \geq N$

$$y[n] = \sum_{k=0}^{N-1} h[n]\,x[n-k]$$

$$= h[0]x[n] + h[1]x[n-1] + \ldots + h[N-1]x[n-N+1]$$

$$y[n] = \frac{1}{2}x[n] + \frac{1}{2}x[n-1] \Rightarrow h[0] = h[1] = 1/2\,,\ h[n] = 0 \quad n \neq 0,1$$

5:55

3) Infinite impulse response (IIR): h[n] has an infinite number of nonzero values

Ex: $$h[n] = \begin{cases} 0 & n<0 \\ (1/2)^n & n \geq 0 \end{cases}$$

Can't compute convolution sum for IIR systems → use difference equation for output

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

$a_k \neq 0$ for $k=1,2,\cdots,N$ $\Rightarrow$ IIR

$a_k = 0$ for $k=1,2,\cdots,N$ $\Rightarrow$ FIR

(use $x[n] = \delta[n]$)

8:14

Example: $y[n] = \frac{1}{2}y[n-1] + x[n]$, Find impulse response $h[n]$

let $x[n] = \delta[n]$: $h[n] = \frac{1}{2}h[n-1] + \delta[n]$

if causal, $h[n] = 0$, $n < 0$

$h[0] = \frac{1}{2}h[-1] + 1 = 1$

$h[1] = \frac{1}{2}h[0] + 0 = \frac{1}{2}$

$h[2] = \frac{1}{2}h[1] + 0 = \frac{1}{4}$

$\vdots$

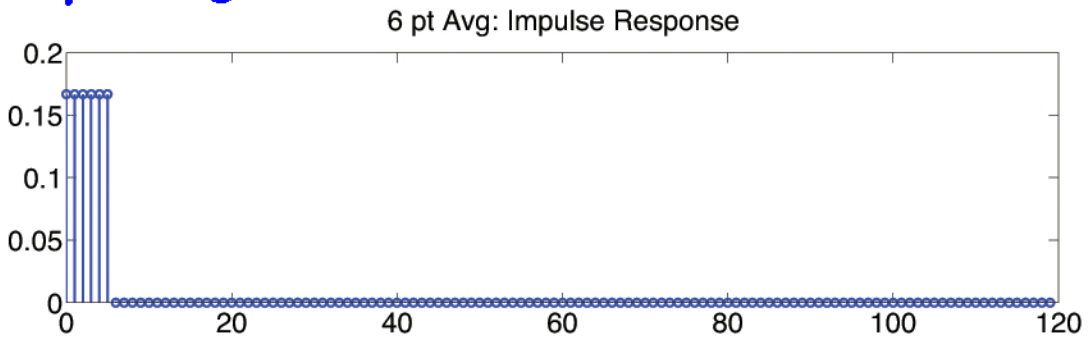
$h[n] = \left(\frac{1}{2}\right)^n$, $n \geq 0$

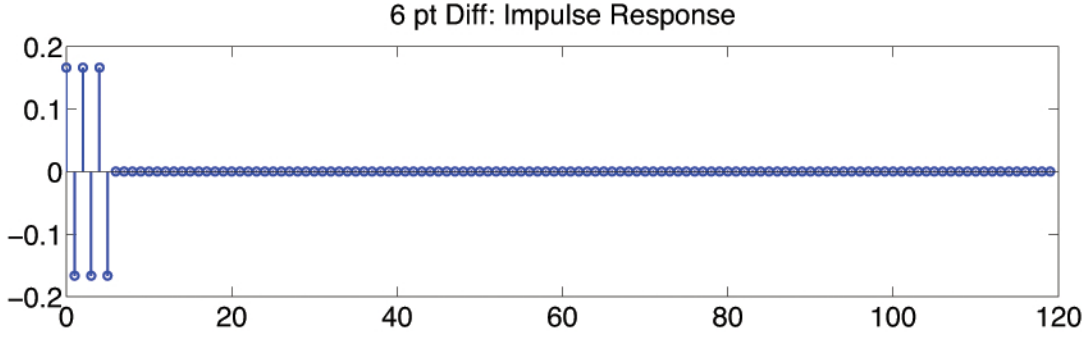
IIR

Examples:

9:33

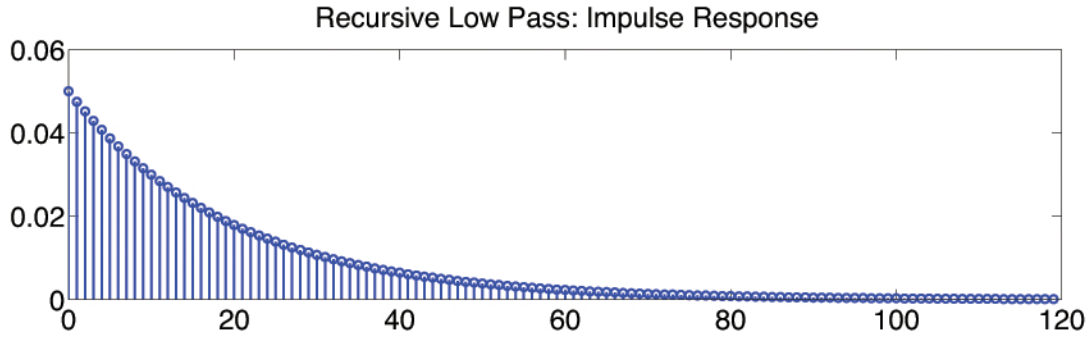
$$y[n] = \frac{1}{6}\{x[n] + x[n-1] + \cdots + x[n-5]\}$$

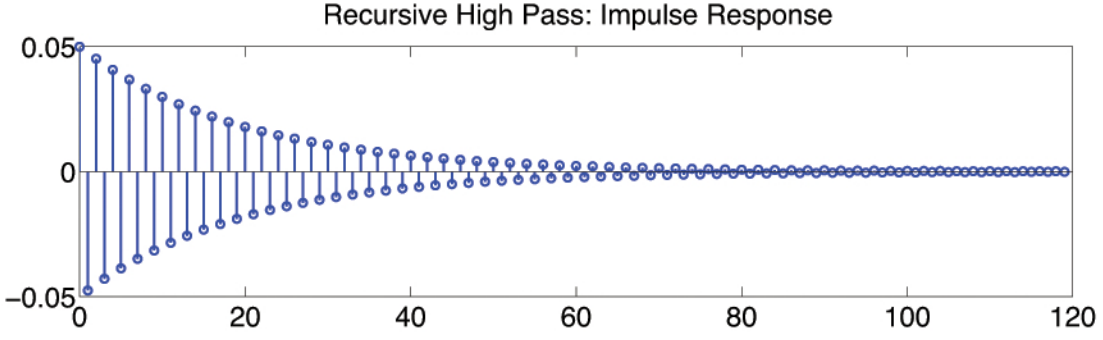




$$y[n] = \frac{1}{6}\{x[n] - x[n-1] + x[n-2] - x[n-3] + x[n-4] - x[n-5]\}$$

$$y[n] = 0.95\,y[n-1] + 0.05\,x[n]$$

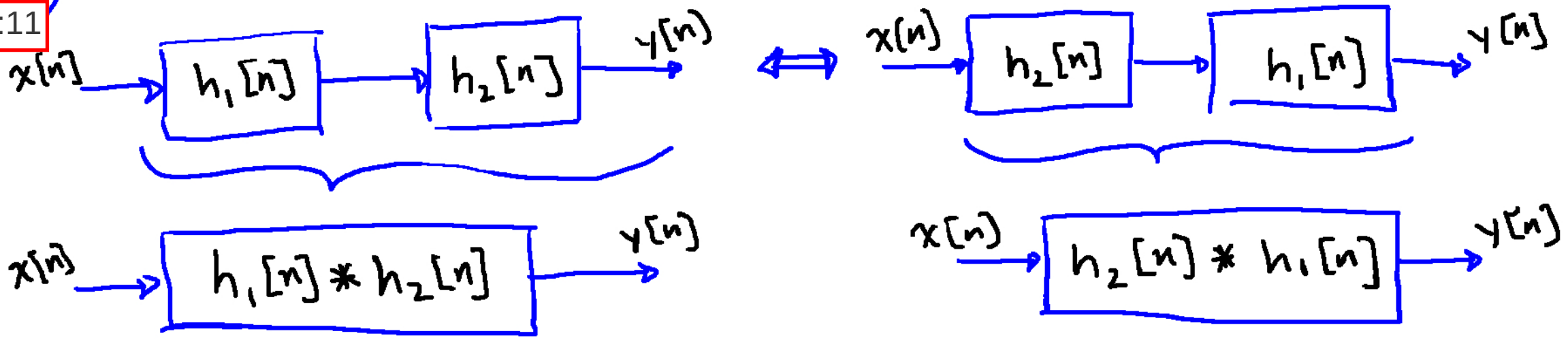




$$y[n] = -0.95\,y[n-1] + 0.05\,x[n]$$

4) Cascade of systems

11:11



$$h_1[n] * h_2[n] = h_2[n] * h_1[n]$$

"commutative"

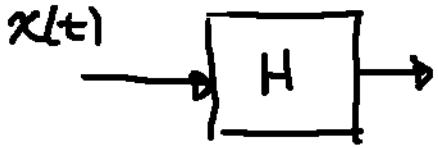
(or $x[n] * h[n] = h[n] * x[n]$)

13:03

Continuous-time systems-



$\Rightarrow$



$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$$= h(t) * x(t)$$

- Impulse response h(t) is a mathematical model for system response to an idealized large amplitude, short duration input
- Impulse response and convolution integral are of value for <u>analysis</u> of continuous-time systems

For more great educational signal-processing content:

AllSignalProcessing.com

- Intuitive, easy-to-follow style
- Organization that saves you time
- Answers to your questions
- Learn at your pace
- Award winning instruction